Jagonmoy Dey

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Research Interests

- Secure Code Generation using LLMs • Software Security • Empirical Software Engineering

Education

B.Sc. in Software Engineering *Jan 2018 – Feb 2023*
Shahjalal University of Science and Technology *Sylhet, Bangladesh*
CGPA: 3.60 / 4.0, Final 4 Semesters: 3.85+/4.00

Research Experience

Volunteer Researcher (Secure Code generation using LLMs) *November 2025 - Present*
Collaborating with Dr. Tanmay Bhowmik (Kennesaw State University) and his research group SERA (Software Engineering Research and Analysis Lab)
DualGuard: A GenAI-Based Automated Approach to Generate Functionally Correct Secure Code
Code available upon request (repo private until research archived/published)

- Reviewed literature on secure code generation and found that prior methods improve security at the cost of functional correctness; formulated this as a research problem and designed and implemented DualGuard end-to-end to solve it.
- Designed DualGuard, a two-stage framework for secure code generation: Stage 1 builds a diverse pool of candidate solutions using multiple generation strategies plus security-aware knowledge retrieval; Stage 2 picks the final solution using a priority order — test-case agreement first, then consensus among candidates, then security score — with no manually written tests needed.
- Benchmarked DualGuard against 4 baselines (RESCUE, PromSec, INDICT, zero-shot) on SecCodePLT+ (1,201 tasks) and HumanEval+, using GPT-4o-mini and Gemini-2.5-Flash-Lite; got the best Secure-Pass@1 of all methods (10.4% vs 8.0–8.4% for baselines, 24–30% higher), using separate CodeQL/Bandit scanners to avoid evaluation leakage.

Research Assistant (Thesis: Speaker Diarization with Speech Recognition in Bengali)
Shahjalal University of Science and Technology July 2022 - Feb 2023
View Thesis | *Python, TensorFlow, ELAN, Audacity*

- Contributed to a thesis on a multi-component speech processing system involving voice activity detection, speaker segmentation, embedding extraction, clustering, and speech recognition.
- Implemented the logic to integrate voice activity detection with speaker segmentation for accurate speaker turn detection, and integrated the Google Speech-to-Text API.
- Awarded **4.00/4.00** for the thesis and defense.

Work Experience

Software Engineer II *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *January 2026 – Present*

- Developing real-time control interfaces for OSC.no, a Norwegian ocean simulation platform, processing high-frequency data streams via NATS pub/sub for remote equipment operation.
- Leveraged Claude Code with agentic workflows, hooks, skills, and MCP integrations to automate repetitive development tasks and streamline engineering pipelines.

Software Engineer I *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *July 2024 – December 2025*

- Built and scaled subrite.no, a multi-tenant SaaS subscription management platform, as a full-stack engineer from initial development through production, serving around 300,000 users.
- Architected an event-driven and scheduled task system for automated email workflows (Nest.js queues), and a content pipeline integrating Sanity CMS with automated type generation, GROQ queries, and templated rendering (MJML, rich text to HTML/PDF).
- Optimized data fetching, caching (Next.js, Redis), and media delivery (MUX, Imgix, Next/Image), raising

Google Lighthouse scores from 55%→95% (performance) and 85%→100% (SEO).

Associate Software Engineer *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *Feb 2023 – June 2024*

- Integrated tenant applications with Subrite's OIDC system via NextAuth, implementing secure authentication flows with dynamic tenant-specific redirect and callback URLs; introduced Playwright end-to-end tests into the CI/CD pipeline alongside Jest unit tests to ensure code reliability.
- Built a multi-tenant frontend with dynamic tenant-specific custom fields, responsive UIs, and reusable component libraries; optimized backend RESTful APIs and TypeORM SQL queries, improving response times by 1.5x.

Trainee Software Engineer *Dhaka, Bangladesh*
Cefalo Bangladesh Limited *Sep 2021 – February 2022*

- Received structured training on Web, HTTP, REST APIs, and software engineering foundations, and implemented these concepts by developing features for Cefalo's internal in-house web platforms.

Projects

AI-Driven Task Reviewer | *Python, FastAPI, ChromaDB* ▶ *View Demo*
Cefalo Bangladesh Limited

- Built an LLM/RAG pipeline over Jira to flag duplicate tasks, incomplete specs, vague summaries, and unattended comments, and generate reports via Slack and email

VocaBuilder - AI-Powered Vocabulary Builder | *Python, FastAPI, LangChain, ChromaDB, React*

- Built semantic retrieval and contextual story-based vocabulary learning powered by GPT-family LLMs.

Time-Series-with-stock | *Python, PyTorch, Scikit-learn, NumPy, Pandas*

- Extended TSLib with support for Kaggle stock datasets and unified model evaluation workflows.

SlackBook | *Javascript, Node.Js, Slack API, Facebook Graph API*

- Built a Slackbot enabling cross-platform message and attachment synchronization with Facebook.

Technical Skills

Languages: Python, JavaScript/TypeScript

Frameworks/Libraries: FastAPI, React.js, Next.js, Nest.js, Express.js, Playwright, Jest, PostgreSQL, Redis

ML/LLM Frameworks: LangChain, TensorFlow

LLM Technologies: ChromaDB, RAG Systems, LLM Integration (OpenAI, Gemini, etc.)

Developer Tools: Git, Docker, Claude, Postman

Competitive Programming

Solved 1200+ problems; participated in 80+ contests; max **Codeforces** rating 1500+; Maintained multiple GitHub repositories for curated data structure and algorithmic resources, such as **Number Theory** and **Graph Theory**, which are used in the department as course materials.

Certifications

Intermediate Front End Secure Coding Learning Path | *Secure coding practices for front-end technologies, covering XSS, CSRF, and CSP* *View Certificate*

Claude Code 101 | *Introduction to agentic coding with Claude Code, covering MCP, subagents, hooks, and skills* *View Certificate*

Language Proficiency

IELTS | Overall: 8.0 (Listening: 9.0, Speaking: 8.5, Reading: 7.0, Writing: 6.5)

Mentorship Experience

- Trainer for Junior Engineers, Cefalo — Led code reviews and workflow sessions.
- Competitive Programming Workshop Instructor, SUST — Conducted algorithm and problem-solving training.